MILITARY SPECIFICATION

MEASURING SET, DEPTH-SOUND SPEED AN/BQH-1 ( )

1. SCOPE

1 1 This specification covers the design, construction and performance characteristics of a completely transistorized depth-sound speed measuring set for use on submarines.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-763 - Steel Bars, Shapes and Forgings—Corrosion Resisting.

MILITARY

MIL-S-867 - Steel Castings, Corrosion Resisting Austenitic.
MIL-R-900 - Rubber Sheets; and Cut, Molded and Extruded Special Shaped Sections - Synthetic, Medium Soft, Low Temperature, Gasket, Application.
MIL-S-901 - Shockproof Equipment, Class HI(High-Impact), Shipboard Application, Tests for.
MIL-C-915 - Cable, Cord and Wire, Electrical (Shipboard Use).
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore General Specification
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts Preparation for Delivery of.
MIL-F-21467 - Fittings, Flareless, Fluid Connection (Shipboard Use).
MIL-E-21981 - Electronics Type Designations, Identification Plates and Markings; Requirements for.
MIL-C-22539 - Connector Sets, Electrical, Hermetically Sealed Submarine.
MIL-R-22732 - Reliability Requirements for Shipboard and Ground Electronic Equipment

STANDARDS

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-415 - Test Points and Test Facilities for Electronic Systems and Associated Equipment, Design Standard for.

DRAWINGS

BUREAU OF SHIPS

RE78D2425 - AN/BQH-1A Measuring Set, Depth-Sound Speed Outline and Mounting Dimensions.
815 1197170 - Connectors Hermetically Sealed for 3, 4 and 5 No. 12 and 8 No. 16 Pins.

PUBLICATIONS

BUREAU OF SHIPS

NAVSHIPS - 93820 - Handbook for the Prediction of Shipboard and Shore Electronic Equipment Reliability.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6625

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Ave., at 33rd Street, New York 16, N. Y.)

3 REQUIREMENTS

3.1 Material.- The material used in the design, fabrication, and construction of this set shall be in accordance with MIL-E-16400, except as otherwise specified herein Particular attention shall be paid to the acceptable and unacceptable material requirements of MIL-E-16400.

3 1 1 Corrosion-resisting materials.- Materials which normally come in contact with sea water shall withstand, if normally maintained, a continuous or intermittent exposure to sea water. Mating or touching surfaces which normally come in contact with sea water shall not be made of dissimilar metals. For this purpose, different grades of corrosion-resistant steel are considered dissimilar metals. Every effort shall be exerted by the contractor to employ only such materials, for this specific purpose, that are known to have a life expectancy in excess of 3 years with regard to their ability to withstand the corrosive effects of sea water or shall have an appropriate external surface treatment or coating which shall be effective in inhibiting corrosion in sea water. The exposed material, the surface treatment, or the coating, will not be acceptable if maintenance services required to keep these parts of the equipment free from harmful corrosion must be performed at intervals of less then 2 years Rubber parts need not conform to the foregoing requirements, but shall be of the best quality, for the intended purpose.

3.1 2 Rubber - Rubber used in the sea unit, if any, shall be in accordance with MIL-R-900 The crystallization temperature or the second order of transition on temperature of the rubber compound shall be below 10 degrees Fahrenheit (°F.), and the rubber shall be compounded for high electrical resistivity. Mold release material used for molding all rubber parts shall not be of the silicone type.

3.2 General requirements.- The measuring set shall meet the requirements of MIL-E-16400, except as otherwise specified herein. The design objectives, as specified in MIL-E-16400 for reliability, simplicity, ease of installation and maintenance consistent with the needs of the Naval service, shall be considered during all phases of the equipment design.

3 2 1 Temperature - The equipment shall meet the temperature requirements of class 4 of MIL-E-16400

3.2.2 Power supply.- The equipment shall be designed to operate from a 115-volt + 10 percent 400 cycle ± 5 percent single-phase power source

3 2 3 Enclosures - Enclosures shall be dripproof in accordance with MIL-STD-108

3 2 4 Maximum size - Units shall not exceed dimensions specified for similar units on RE-78-D-2425.

3 2.4.1 Accessibility.- The equipment shall be designed mechanically so that it can be serviced from the front, and no clearance is required beyond the planes defined by the sides of the cabinet.

3.2.5 Installed weight - The installed weight of the complete equipment shall be kept to a minimum commensurate with good engineering design Installed weight shall include all major units and essential accessories thereof necessary to make such equipment fully operative in the system

3 2.6 Special cables and fittings - All special cables and fittings shall be supplied by the contractor. Cables used or specified for interconnection purposes shall be in accordance with MIL-C-915.

3.2 7 Extra terminals.- Extra terminals shall be in accordance with MIL-E-16400, unless specifically approved by the bureau or agency concerned

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3.2.8 Service conditions.- The equipment shall be designed to give optimum performance and reliable service during continuous or intermittent operating periods under all conditions likely to be encountered in Navy service without the equipment overheating or the necessity for the major adjustments or servicings.

3.2.9 Operable time - The equipment shall be fully operable within 3 minutes after the main power is turned on.

3.2.10 Operating controls - Arrangement of all operating controls shall be in accordance with MIL-E-16400. Approval of the human engineering factors involved shall be obtained from the bureau or agency concerned before making the final layout firm. No locking devices shall be required on controls

3.2.11 Transistors.- The equipment shall be completely transistorized. Germanium transistors or diodes shall not be used.

3.2.12 Reliability.- The reliability goal for this equipment shall be 1500 hours, mean-time-between-failures, (MTBF), with a confidence level of 90 percent. The MTBF of the equipment shall be predicted using method D of NAVSHIPS 93820. For test purposes, demonstration of attainment of a reliability level of 300 hours MTBF with a 90 percent confidence level only will be required.

3.2.13 Shock.- The equipment shall be capable of withstanding the shock requirements for type A, light weight equipment, of MIL-S-901.

3.3 Equipment function.- The basic function of the measuring set is to continuously measure and record sound speed in water as a function of depth. The equipment shall automatically function in the following manner:

(a) Transmit an acoustic pulse, thru the water, over a known path length.
(b) Receive and measure the time interval between the transmission and reception of the pulse.
(c) Automatically convert the time measurement into a sound speed measurement in feet per second and record the information on a chart.
(d) Measure water pressure.
(e) Automatically convert the water pressure measurement into a depth measurement in feet and record the information on a chart.

3.4 Operating limits.- The equipment shall be capable of measuring and recording sound speed over a range of 4600 to 5100 feet per second and depth from 0 to 1500 feet. Two overlapping depth scales, 0 to 800 feet and 700 to 1500 feet shall be provided.

3.5 Accuracy.- The equipment shall not deviate more than 0.03 percent from the temperature-sound velocity information furnished on figure 1 (see 3.6.3.5.1).

3.6 Major units.- The equipment shall consist of the following major units.

(a) Recorder.
(b) Remote recorder-repeater.
(c) Sea unit (two required).
(c) Depth element.

3.6.1 Recorder.- The recorder shall be servo-controlled and shall consist of a chart, a chart drum and driving mechanism, a recording stylus, a stylus driving mechanism and all necessary control circuits for indicating sound speed and depth as a function of the output signals of the sea unit and depth element. The recorder shall be capable of furnishing synchro generated output signals, indicative of the position of the drum and stylus, to the remote-recorder repeater. Power supplies for the depth element and sea unit shall be contained in the recorder housing.

3.6.1.1 Controls.- The following controls shall be provided on the front panel:

(a) Sea unit selector switch.- A selector switch shall be provided on the front panel to switch between the upper and lower sea units and shall operate to automatically shift, the depth stylus to obtain true sea unit depth recording The indicated depth of each sea unit shall be adjustable to compensate for differences between various classes of submarines and shall have an adjustment range of not less than 50 feet. A locking device shall be provided to permit

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securing the adjustment at the time of installation. Indicator lights shall be provided to show which sea unit is in use.

(b) Depth scale select switch. - The depth presentation shall be in two ranges.
(1) 0 to 800 feet.
(2) 700 to 1500 feet.
A selector switch shall be provided on the front panel to select either depth range. Indicator lights shall be provided to indicate the range in use.

(c) Power switch - A power-on-off switch shall be provided on the front panel. This switch shall disconnect power to all units.

(d) Chart calibration - A spring-loaded toggle switch shall activate the sea unit calibration circuits.

3.6.1.2 Illumination - Red illumination using red lamps shall be provided The intensity of this illumination shall be variable from "full" to "off" by means of a manual control.

3.6.1.3 Resolution. - The recorder resolution shall be sufficient to enable reading changes in the speed of sound of 1 foot per second.

3.6.1 4 Response. - The recorder response shall be critically damped for both velocity and depth and shall read zero to full scale in six seconds.

3.6 1.5 Chart - The recorder chart shall be of the pressure sensitive paper type The chart shall be readily replaceable, and access thereto shall be by means of a hinged door on the front panel. The door shall contain a transparent lucite window amber 2451, or equal, to allow viewing of the chart. Chart format shall be as shown on figure 2 Chart calibration for sound speed shall be on the long axis and shall cover the range of 4600 to 5100 feet per second in increments as shown in figure 2. Depth calibration shall be 0 to 800 feet and 700 to 1500 feet in increments as shown on figure 2. The number of charts to be supplied shall be specified (see 6.1).

3.6 1.6 Isoballast curves. - A family of curves shall be printed on each chart such that each curve passes through all the points where the loss in buoyancy with depth due to compression is exactly equal to the gain in buoyancy due to a decrease in temperature or increase in salinity. The area on the chart between sound velocity of 4725 and 4776 feet per second shall contain no curves. The family of curves to the right of this area shall be of the isosaline water case and to the left shall be the isothermal case. Isoballast curves shall be calculated and printed for the classes of submarines specified (see 6.1). Information concerning submerged displacement and hull compressibility required for the above calculations and spacing between the curves will be forwarded upon request by contractor to the bureau or agency concerned (A more detailed discussion on isoballast curves and submarine buoyancy control may be found in Summary of Submarine Technical Report of Division 6, NDRC, Volume 6B entitled Methods of Submarine Buoyancy Control. A copy of the study may be obtained by the contractor upon request to the bureau or agency concerned)

3.6.1.7 Recorder cabinet - The recorder cabinet shall be designed for bulkhead mounting, without shockmounts. The dimensions and weight of the recorder shall not exceed the following

(a)	Height	13-5/8 inches
(b)	Width	12-3/16 inches
(c)	Depth	9-7/8 inches
(d)	Weight	50 pounds

3.6.2 Remote recorder repeater. - The remote recorder repeater shall accept signals and control voltages from the recorder, and shall provide a remote display of sound speed versus depth. The recorder mechanism and display shall meet the requirements specified in 3.6.1. Sound velocity displayed on this unit shall not deviate more than 0.06 percent from the temperature-sound velocity data shown on figure 1.

3 6.2.1 Controls. - The following operating controls shall be provided on the front panel

(a) A power switch and power-on indicator.
(b) An illumination intensity control.
(c) An indication of which sea unit is in use.
(d) An indication of which depth range is in use.

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3.6.2.2 All gears or couplings associated with mechanical assemblies shall be pinned or keyed to the shafts.

3.6.2.3 Test points.- Unless otherwise specified herein, test points and test features shall be in accordance with MIL-STD-415

3.6.2.4 Elapsed time meter.- An elapsed time meter in accordance with MIL-E-16400 shall be provided to give a continuous record of the elapsed operating time of this equipment.

3.6.2.5 No blown fuse indicators shall be furnished.

3.6.3 Sea unit.- The sea unit shall consist of a pair of transducers and reflector plates fastened to a plate to form a sound path of fixed length. This plate shall be mounted on a housing containing the necessary electronic circuits (see figure 3).

3.6.3.1 Output - The sea unit shall furnish a direct current (d.c.) signal which varies from 3.9 volts at 4600 feet per second to 9 volts at 5100 feet per second in exact proportion to the change in sound speed. Provision shall be made to permit calibration of the output signal at the recorder. Each sea unit shall have a calibration identity which will calibrate the unit at 3 points (low, medium, and high sound speed) along its scale so as to reproduce the temperature-velocity scale within the required accuracy (see 3.5).

3.6.3.2 Interchangeability.- The sea units shall be interchangeable and useable with any recorder. At any given temperature the sound velocity indication shall not deviate more than plus or minus 7 feet per second from any other sea unit prior to any calibration at the recorder.

3.6 3.3 Transducers.- The transmitting and receiving transducers shall be electrically and mechanically interchangeable. Dowel pin holes shall be provided to accurately position the transducers so that all replacement transducers shall be interchangeable within plus or minus 2 feet per second. The transducers shall have a frequency response in excess of 1 megacycle to properly reproduce and transmit the high frequency transient pulses required for proper operation The electrical and mechanical "Q" shall be no greater than 4 so that the total transient time does not exceed 4 microseconds for the total signal when transmitted and received through the transducer system.

3.6.3.3.1 Pressure effect - When each sea unit is subjected to a pressure of 1000 pounds per square inch (p.s.i.) the recorder shall indicate an increase of 39 feet per second plus or minus 2 feet in sound speed.

3.6.3.4 Transmitter - The transmitter shall produce a high frequency electrical pulse whose period is less than half a microsecond. The transmitter when loaded by the radiating transducer shall produce a highly damped pulse so that the total ringing time of transducer and transmitter shall be less than 3 microseconds. The acoustic pulse shall travel over a path length of at least 8 inches to the receiving transducer.

3.6.3.5 Amplifier.- The received signal shall drive a wide band amplifier whose output is thresholded and suitably gated to properly drive the output trigger circuit. The amplifier shall have associated with it a remote control relay which inserts a calibration signal into the amplifier chain.

3.6.3.5.1 Calibration circuit - The sea unit shall contain sound velocity calibration circuits (see 3.6.3.5). The calibration circuits shall provide an accurate time delay signal which will simulate a 4800 foot per second sound velocity. This delay shall not drift more than plus or minus 0.5 foot per second (see 3.6.3.17). After the chart paper has been positioned by use of the calibrating circuit, the entire velocity scale shall track the data shown on figure 1 with an error not greater than 0.03 percent.

3.6.3.6 Oscillator.- A crystal controlled oscillator shall be used to establish the timing pulses and the repetition rate of the transmitted pulses. The output of the oscillator shall be differentiated to provide accurate, jitter free control of the amplifier and the on or off trigger circuits. Oscillator drift shall be less than 0.01 percent over the operating temperature range (see 3.6.3.17).

3.6.3.7 Power supply.- The sea unit power supply shall be housed in the recorder. The following voltages shall be provided:

(a) 28-volt d.c. series regulated power source.
(b) 12-volt d.c low impedance emitter follower power source for the clamping voltage supply.

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3.6.3.8 Parts.- Care shall be taken in the selection of sea unit parts to insure extreme stability over protracted periods of operation.

3.6.3.9 Dimensions and weight.- The dimensions and weight of the sea unit shall not exceed the following:

(a)	Diameter	7-7/8 inches including mounting flange.
(b)	Length	9 inches.
(c)	Weight	55 pounds.

3.6.3.10 Housing.- The sea unit housing and associated metallic parts shall be made of corrosion-resistant steel conforming to Class 316 of QQ-S-763. Corrosion-resistant steel casting, if used, shall conform to Class III of MIL-S-867.

3.6.311 Transducer mounting plate.- The transducer mounting plate shall be fabricated of corrosion-resistant steel conforming to Class 316 of QQ-S-763. The reflectors shall be welded to the mounting plate. Dowel pins shall be provided to accurately position the transducers.

3.6.3.12 Cable.- Each sea unit shall be supplied with 125 feet of watertight cable, type 7SS2.

3.6.3.13 Seal.- Cable shall be connected to the sea unit by a hermetically sealed connector plug and receptacle in accordance with Drawing 815-1197170 and MIL-C-22539.

3.6.3.14 Hydrostatic pressure.- Each sea unit and its cable shall be capable of withstanding repeated subjections to a hydrostatic pressure of 1000 (p.s.i.g.) without physical or electrical damage, or without causing leakage of sea water into the sea unit or connector (see 4.5.1)

3.6.3.15 Hydraulic shock.- The sea unit shall be designed to withstand shock of 7 5 p.s.i. seconds, superimposed on a hydrostatic pressure of 578 p.s.i. Because of short time duration of the impulse shock, peak pressure of approximately 7,500 p.s.i. shall be applied to the unit

3.6.3.16 Pull - The cable shall withstand a 50 pound pull without damage to the seal or without the cable pulling out when tested as specified (see 4.5.2).

3.6.3.17 Operation, sea water.- The sea unit shall operate under the requirements specified herein, at any temperature between 28° and 90° F. This sea unit shall be designed to withstand continuous salt water exposure.

3.6.3.18 Marking for identification.- Terminals on the terminal board of the sea unit, and the external electrical leads, shall be suitably and clearly marked so that positive identification of each lead and terminal can be made by simple visual examination.

3.6.3.19 Grounding of leads.- Electrical circuits shall not be grounded to the sea unit housing.

3.6 3 20 Inspection and repair.- The construction and assembly of the sea unit shall be such that it can be opened and disassembled for inspection and repairs without the destruction of parts except gaskets and packing, and without the use of techniques or tools which are not ordinarily available in the shops of a Naval base

3.6.4 Depth element - The depth element shall be of the variable reluctance type and produce an output signal proportional to the pressure applied. The depth element shall operate over the range of 0 to 750 p.s.i. and withstand pressures of 1200 p.s.i. without damage or degradation of performance. All electronic and magnetic units shall be isolated from sea water.

3.6 4.1 Power supply.- The depth element shall be designed to operate from a 7.5 volt root mean square (r.m.s.), 400 cycle power supply located in the recorder cabinet.

3.6 4.2 Sensitivity.- The depth element shall have a sensitivity of 50 millivolts per volt of applied excitation for an indicated 1500-foot depth.

3 6 4.3 Linearity. The depth element, recorder and repeater, shall be linear within plus or minus 0.5 percent of full scale depth

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3.6.4.4 Hysteresis.- Hysteresis of the depth element and recorder shall not exceed plus or minus 0.1 percent of full scale depth. Hysteresis of the repeater shall not exceed plus or minus 0.15 percent of full scale depth.

3.6.4.5 Resolution.- The resolution of the depth element shall be such that a change in depth of 6 inches shall produce measurable change in balance of the depth element at any depth.

3.6.4.6 Pressure connection.- A 1/4-inch tube size, male, flareless fitting conforming to MIL-F-21467 shall be provided to connect the depth element to the submarine's internal pressure line

3.6.4.7 Material.- All metallic parts of the depth element subject to immersion in sea water shall be of corrosion-resistant steel conforming to Class 316 of QQ-S-763.

3.6.4.8 Accuracy.- On the 0-800 foot scale, the absolute error shall be no greater than 2.5 feet. On the 700-1500 foot scale, the absolute error shall be no greater than 5 feet. Absolute depth shall be determined by a dead weight tester using 44.5 pounds per square inch pressure for 100 feet of length.

3.7 Nomenclature and marking.-

3.7.1 Nomenclature.- Nomenclature shall be requested in accordance with MIL-E-21981.

3.7.2 Serial numbers.- Information on serial numbers shall be obtained from the bureau or agency concerned

3.7.3 Marking.- Each transducer assembly shall be marked with the appropriate nomenclature identification and serial number. These shall be permanently affixed on the transducer assembly by a method satisfactory to the bureau or agency concerned. The marking shall be at the cable end of the assembly unless otherwise specified in the contract or order.

3.7.3.1 Each production transducer assembly offered for delivery shall be accompanied by a data sheet containing pertinent information. The manufacturer shall submit a proposed data sheet for approval by the government.

3.8 Preproduction sample.- Prior to beginning production a sample shall be tested as specified in 4.3 (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General inspection.-

4.2.1 Classification of inspection.- The methods of examination and listing shall be as follows:

(a) Preproduction inspection.
(b) Quality conformance inspection.

4.3 Preproduction inspection.- Preproduction inspection shall consist of the examination and tests specified in 4.3.1.

4.3.1 Preproduction inspection.- Preproduction inspection shall be performed on a complete prototype model at a laboratory satisfactory to the bureau or agency concerned. Evaluation data and sketches or drawings showing the detail construction shall be furnished with the model. These tests shall be performed in accordance with the requirements of MIL-E-16400 and shall consist of any examination or tests considered necessary to determine conformance with this specification. Salt spray test is not required.

4.4 Quality conformance inspection -

(a) Production inspection.
(b) Quality conformance control inspection.

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4.4.1 Production inspection.- Production inspection shall be made at the place of manufacture. Unless otherwise specified by the bureau or agency concerned, production inspection shall be made on all units and shall include the production inspection of MIL-E-16400 and the tests specified herein.

4.4.2 Production control inspection.- Production control inspection shall be in accordance with MIL-E-16400. Production control inspection shall be conducted on one out of each 25 production equipments or fraction thereof Equipment shall be selected at random

4.5 List of tests and test procedures.- A list of proposed test and test procedures shall be furnished by the contractor to the bureau or agency concerned for approval prior to setting up test equipment for the contractor's inspection and tests.

4.5.1 Pressure cycling test.- Each sea unit and depth element (includes unit, gland and cable outside pressure hull) shall be tested to comply with 3.6.3.14 The pressure test shall be applied for 25 cycles with the pressure starting from zero p s.i. and building up to 1000 p.s.i.g maximum during each cycle. The maximum pressure shall be maintained for 1 minute during the first cycle and for 5 to 10 seconds during subsequent cycles. During and after the above tests, there shall be no evidence of physical or electrical damage, or leakage of sea water into the sea unit.

4.5.2 Pull test - The cable shall be tested to determine conformance with 3.6.3.16. The pull test may be conducted as follows With the sea unit clamped rigidly and the cable wound five turns around a 3-inch diameter cylinder, a 25 pound yoke pull applying a load equally to both ends of the cylinder shall be applied axially to the sea unit until a scale measuring the full force registers the amount specified herein Force shall be applied five times holding at maximum pull each time for 1 minute.

4.5 3 Reliability testing.- Reliability testing shall be performed as specified in MIL-R-22732 for temperature and humidity, vibration testing (vibration testing may be performed independently of other environmental conditions) and on off cycling and input voltage.

4.5.3.1 Accelerated life tests specified in MIL-E-16400 will not be required In lieu of this test reliability tests as specified herein shall be required.

4 5.3.2 Demonstration of a minimum inherent reliability level of 300 hours MTBF will be considered as demonstration of compliance with the reliability requirements specified herein. The minimum test time on each equipment shall be no less than 450 hours nor more than 700 hours. Accept, reject criteria shall be in accordance with the accept reject criteria specified in MIL-R-22732. If an accept decision cannot be made before 700 hours the test shall be terminated and the required reports of the tests as specified in MIL-R-22732 shall be submitted to the bureau or agency concerned The decision to accept or reject will be made by the bureau or agency concerned based on the test reports.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts -

5.1 1 Basic equipment or item -

5 1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the suppliers commercial practice when such meets these requirements

5.1.1.2 Packing - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1 3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

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5.1.2 Repair parts.- Repair parts shall be provided as specified.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity (see 6.1).

5 2.1.1 Preservation packaging, packing and marking.- The equipment, and repair parts shall be prepared for domestic shipment-immediate use, domestic shipment and storage or overseas shipment as specified in the contract or order in accordance with MIL-E-17555.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of charts required (see 3.6.1.5).
(c) Classes of submarines for which isoballast lines are desired (see 3.6.16)
(d) Preservation packaging, packing or marking requirements other than those required by paragraph 5.1 (see 5.2).

6.2 Preproduction.-

6.2.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Patent notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity.
Navy - Ships
(Project 6625-N261Sh)

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Figure 1- Velocity of Sound (ft/sec) In Distilled Water
vs.
Temperature (°C)

Temp. °C	Velocity ft/sec	Temp. °C	Velocity ft/sec
0	4602.11	5.2	4682.97
0.1	4603.74	5.3	4684.42
0 2	4605.37	5.4	4685.87
0.3	4607.00	5 5	4687.33
0.4	4608.63	5 6	4688.78
0 5	4610 26	5.7	4690.23
0.6	4611.89	5.8	4691 69
0.7	4613 52	5.9	4693.14
0.8	4615.15	6.0	4694.56
0.9	4616.79	6.1	4695.98
1.0	4618.42	6.2	4697.39
1.1	4620.01	6.3	4698.81
1.2	4621.61	6.4	4700.23
1.3	4623.20	6.5	4701.65
1 4	4624.79	6.6	4703.06
1.5	4626.32	6.7	4704.48
1 6	4627.99	6.8	4705.90
1.7	4629.58	6.9	4707.32
1.8	4631.18	7.0	4708.74
1 9	4632.77	7.1	4710 12
2.0	4634.36	7.2	4711.51
2.1	4635 92	7.3	4712.89
2.2	4637.48	7.4	4714.28
2.3	4639.04	7.5	4715.66
2.4	4640.59	7.6	4717.05
2.5	4642.15	7.7	4718.43
2.6	4643.71	7.8	4719.82
2.7	4645.27	7.9	4721.20
2.8	4646 83	8.0	4722.58
2.9	4648.39	8.1	4723.93
3 0	4649.94	8 2	4725.28
3 1	4651 46	8.3	4726.64
3.2	4652.98	8.4	4727.99
3.3	4654.51	8.5	4729.34
3.4	4656 03	8.6	4730.69
3 5	4657.55	8.7	4732.04
3 6	4659.07	6.8	4733 39
3.7	4660.60	8.9	4734 75
3.8	4662.12	9.0	4736.10
3.9	4663.64	9 1	4737.42
4.0	4665.16	9.2	4738.74
4.1	4666.65	9.3	4740.05
4 2	4668.14	9.4	4741 37
4.3	4669.62	9.5	4742 69
4.4	4671.11	9.6	4744.01
4.5	4672.59	9.7	4745.33
4.6	4674.08	9.8	4746 65
4.7	4675.57	9.9	4747.97
4.8	4677.05	10.0	4749.25
4.9	4677.54	10.1	4750.54
5.0	4680.06	10.2	4751.83
5.1	4681.51	10.3	4753.11

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Figure 1- Velocity of Sound (ft/sec) In Distilled Water
vs.
Temperature (°C) (Continued)

Temp. °C	Velocity ft/sec	Temp. °C	Velocity ft/sec
10.4	4754.40	15.6	4817.28
10.5	4755.68	15.7	4818.42
10.6	4756.97	15.8	4819.55
10.7	4758.26	15.9	4820.69
10.8	4759.54	16.0	4821.83
10.9	4760.83	16.1	4822.93
11.0	4762.11	16.2	4824.04
11.1	4763.37	16.3	4825.14
11.2	4764.62	16.4	4826.25
11.3	4765.87	16.5	4827.35
11.4	4767.13	16.6	4828.46
11.5	4768.38	16.7	4829.56
11.6	4769.63	16.8	4830.66
11.7	4770.89	16.9	4831.78
11.8	4772.14	17.0	4832.85
11.9	4773.39	17.1	4833.92
12.0	4774.68	17.2	4835.00
12.1	4775.90	17.3	4836.08
12.2	4777.13	17.4	4837.15
12.3	4778.35	17.5	4838.23
12.4	4779.57	17.6	4839.30
12.5	4780.80	17.7	4840.38
12.6	4782.02	17.8	4841.46
12.7	4783.25	17.9	4842.53
12.8	4784.47	18.0	4843.61
12.9	4785.69	18.1	4844.66
13.0	4786.92	18.2	4845.70
13.1	4788.11	18.3	4846.75
13.2	4789.30	18.4	4847.80
13.3	4790.50	18.5	4848.84
13.4	4791.69	18.6	4849.89
13.5	4792.89	18.7	4850.94
13.6	4794.08	18.8	4851.98
13.7	4795.28	18.9	4853.03
13.8	4796.47	19.0	4854.08
13.9	4797.66	19.1	4855.10
14.0	4798.83	19.2	4856.12
14.1	4799.99	19.3	4857.14
14.2	4801.16	19.4	4858.16
14.3	4802.32	19.5	4859.18
14.4	4803.48	19.6	4860.20
14.5	4804.65	19.7	4861.22
14.6	4805.81	19.8	4862.24
14.7	4806.98	19.9	4863.26
14.8	4808.14	20.0	4864.31
14.9	4809.31	20.1	4865.31
15.0	4810.47	20.2	4866.30
15.1	4811.61	20.3	4867.79
15.2	4812.74	20.4	4868.29
15.3	4813.88	20.5	4869.28
15.4	4815.01	20.6	4870.28
15.5	4816.15	20.7	4871.27

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Figure 1- Velocity of Sound (ft/sec) In Distilled Water
vs.
Temperature (°C) (Continued)

Temp. °C	Velocity ft/sec	Temp. °C	Velocity ft/sec
20.8	4872 26	26.0	4920.02
20.9	4873 26	26 1	4920.86
21.0	4874.25	26 2	4921.70
21.1	4875 22	26 3	4922 54
21.2	4876.19	26.4	4923.38
21.3	4877 16	26.5	4924.22
21.4	4878 12	26.6	4925.06
21 5	4879.09	26.7	4925.90
21.6	4880.06	26.8	4926.74
21.7	4881 03	26.9	4927.58
21.8	4881.99	27.0	4928.42
21.9	4382.96	27 1	4929.23
22.0	4883 90	27.2	4930 05
22.1	4884.84	27.3	4930.87
22 2	4885 78	27.4	4931.69
22 3	4886.72	27.5	4932.50
22.4	4887.66	27.6	4933.32
22 5	4888.61	27.7	4934.14
22.6	4889.55	27.8	4934 95
22 7	4890 49	27 9	4935.77
22 8	4891.43	28.0	4936.55
22 9	4892.37	28.1	4937.34
23.0	4893 31	28.2	4938 14
23 1	4894 23	28 3	4938.93
23.2	4895 14	28 4	4939.72
23.3	4896 06	28 5	4940.51
23.4	4896 97	28.6	4941.30
23.5	4897.89	28 7	4942 09
23 6	4898.81	28.8	4942.88
23 7	4899 72	28 9	4943.67
23.8	4900 64	29 0	4944 49
23.9	4901 55	29.1	4945.26
24.0	4902.47	29.2	4946.03
24.1	4903.36	29.3	4946.80
24.2	4904.24	29.4	4947.56
24.3	4905.13	29.5	4948.33
24.4	4906.02	29.6	4949.10
24.5	4906 91	29.7	4949.87
24.6	4907.80	29.8	4950.64
24.7	4908.69	29.9	4951.40
24.8	4909.58	30.0	4952.17
24 9	4910 47	30.1	4952.92
25.0	4911.36	30.2	4953.66
25.1	4912.22	30.3	4954.40
25.2	4913.09	30.4	4955.15
25.3	4913.96	30.5	4955.89
25.4	4914.82	30.6	4956.64
25.5	4915.69	30.7	4957.38
25.6	4916 55	30.8	4958.13
25.7	4917.42	30.9	4958.87
25.8	4918.29	31.0	4959.62
25.9	4919.15	31.1	4960.34

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Figure 1- Velocity of Sound (ft/sec) In Distilled Water
vs.
Temperature (°C) (Continued)

Temp. °C	Velocity ft/sec	Temp. °C	Velocity ft/sec
31.2	4961.06	36.4	4996.03
31.3	4961.78	36.5	4996.64
31.4	4962.51	36.6	4997.26
31.5	4963.23	36.7	4997.87
31.6	4963.95	36.8	4998.48
31.7	4964.67	36.9	4999.10
31.8	4965.39	37.0	4999.71
31.9	4966.11	37.1	5000.30
32.0	4966.84	37.2	5000.90
32.1	4967.54	37.3	5001.49
32.2	4968.24	37.4	5002.08
32.3	4968.94	37.5	5002.68
32.4	4969.64	37.6	5003.27
32.5	4970.35	37.7	5003.87
32 6	4971.05	37.8	5004.46
32 7	4971.75	37.9	5005.05
32.8	4972.45	38.0	5005.65
32 9	4973.15	38.1	5006.22
33.0	4973.86	38.2	5006.80
33.1	4974.54	38.3	5007.37
33.2	4975.22	38.4	5007.94
33.3	4975.89	38.5	5008 52
33.4	4976.57	38.6	5009.09
33.5	4977.25	38.7	5009.67
33.6	4977.93	38.8	5010.24
33.7	4978.61	38.9	5010.81
33.8	4979.29	39.0	5011.39
33 9	4979.97	39.1	5011.94
34.0	4980.65	39.2	5012.50
34 1	4981.30	39.3	5013.05
34.2	4981.96	39.4	5013.61
34.3	4982.62	39.5	5014.16
34.4	4983.27	39.6	5014.72
34 5	4983.93	39.7	5015 27
34.6	4984.59	39.8	5015.83
34.7	4985.24	39.9	5016.38
34.8	4985 90	40.0	5016.93
34.9	4986.55	40.1	5017.47
35.0	4987.21	40 2	5018.00
35 1	4987.85	40.3	5018.54
35.2	4988.48	40.4	5019.07
35.3	4989.12	40.5	5019.61
35.4	4989.76	40.6	5020.14
35.5	4990.39	40.7	5020.68
35.6	4991.03	40.8	5021.21
35.7	4991.67	40.9	5021.75
35.8	4992.30	41.0	5022.25
35.9	4992.94	41.1	5022.76
36.0	4993.57	41.2	5023.28
36.1	4994.19	41.3	5023.79
36.2	4994.80	41.4	5024.31
36.3	4995.41	41.5	5024.82

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Figure 1- Velocity of Sound (ft/sec) In Distilled Water
vs.
Temperature (°C) (Continued)

Temp. °C	Velocity ft/sec	Temp. °C	Velocity ft/sec
41.6	5025.34	44.4	5038.93
41.7	5025.85	44.5	5039.39
41.8	5026.37	44.6	5039.85
41.9	5026.88	44.7	5040.30
42 0	5027.40	44.8	5040.76
42.1	5027.89	44 9	5041 21
42.2	5028.39	45.0	5041.67
42.3	5028.89	45.1	5042.11
42 4	5029.38	45.2	5042.55
42.5	5029.88	45 3	5042 99
42 6	5030.37	45 4	5043.43
42 7	5030.87	45.5	5043.87
42.8	5031.36	45.6	5044.31
42 9	5031.86	45.7	5044.75
43.0	5032.35	45.8	5045.19
43 1	5032.83	45.9	5045.63
43.2	5033.30	46.0	5046 07
43.3	5033.78	46.1	5046.49
43 4	5034.26	46.2	5046.91
43.5	5034.73	46.3	5047.33
43.6	5035.21	46 4	5047.75
43.7	5035.68	46.5	5048.17
43.8	5036.16	46.6	5048.59
43.9	5036.63	46.7	5049.01
44.0	5037.11	46.8	5049.43
44.1	5037.57	46 9	5049.85
44.2	5038.02	47.0	5050.27
44.3	5038.48		

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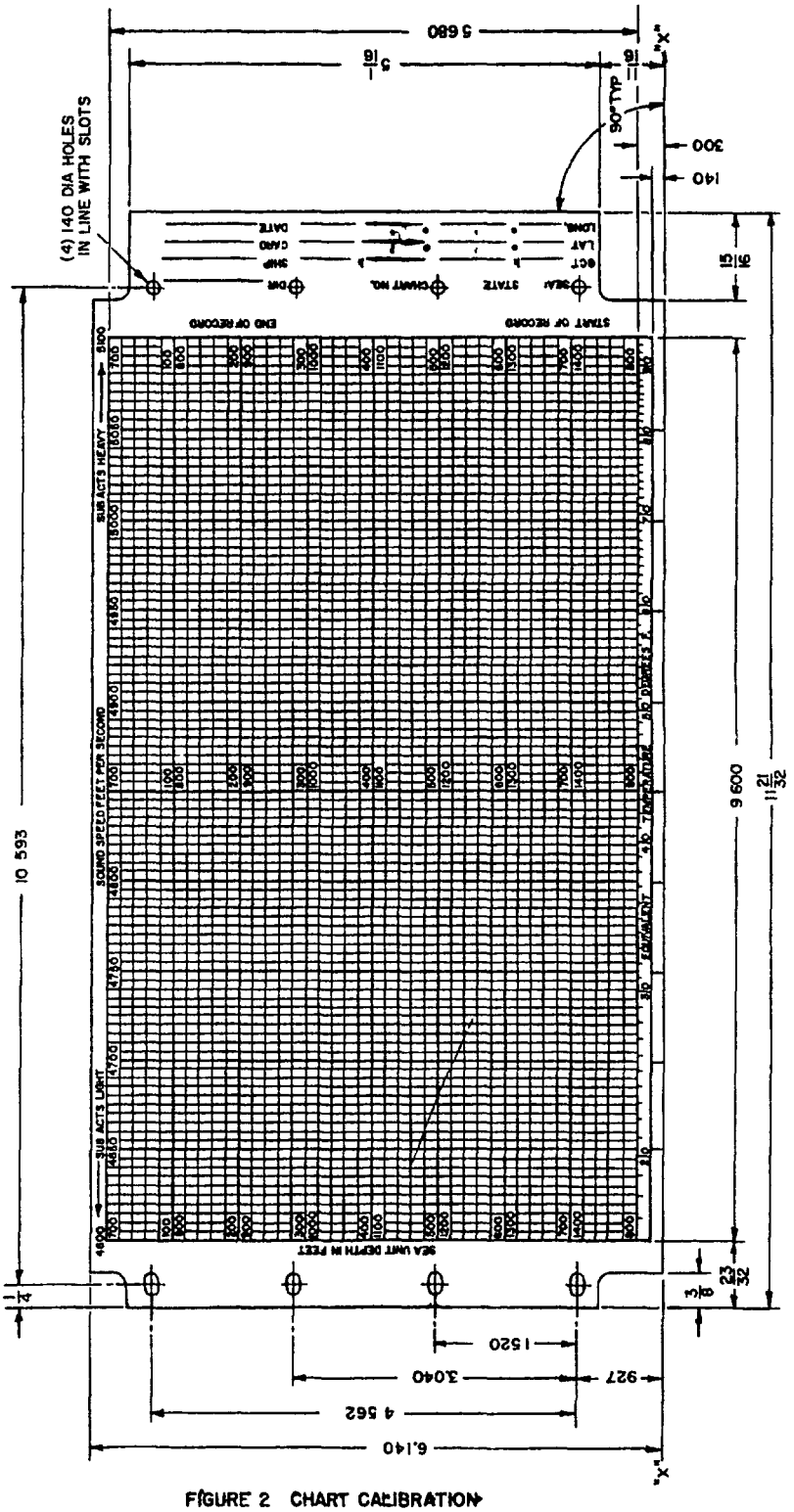
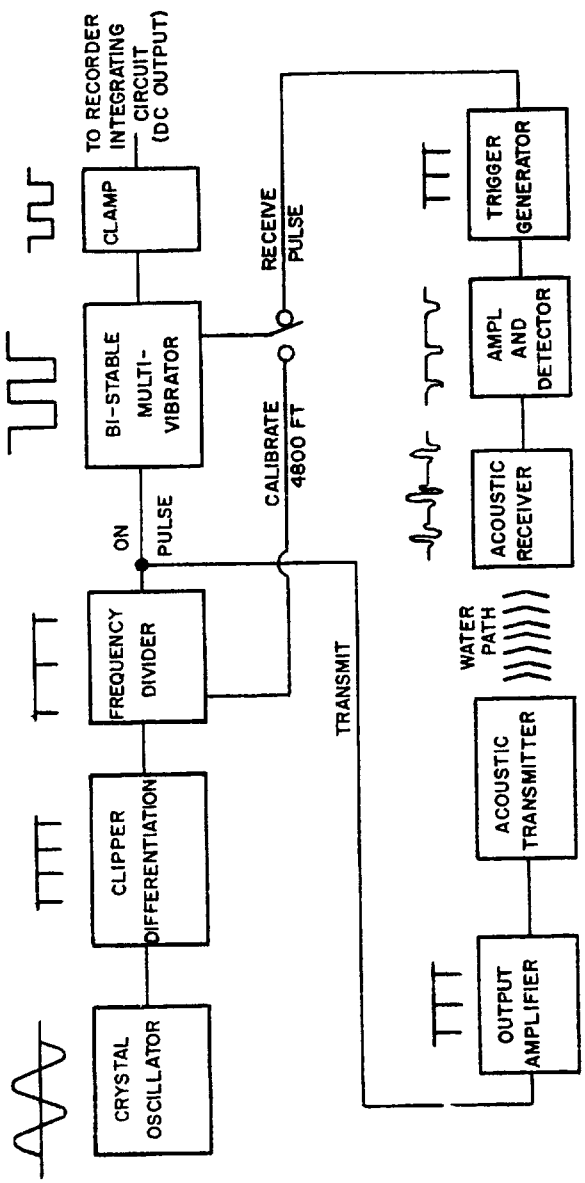


FIGURE 2 CHART CALIBRATION

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FIGURE 3 SEA UNIT BLOCK DIAGRAM

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a. GIVE PARAGRAPH NUMBER AND WORDING

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3 IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

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